

Half-degree irreducibles over $\mathbb{F}_2$: what is proved and what remains open

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Status. This note does not prove Lemire's conjecture. The estimate shown in red in Figure 1 is still open.

Abstract

Barrett reduction over $\mathbb{F}_2$ is especially cheap when an irreducible polynomial f of degree n satisfies $\deg(f - x^n) \leq \lfloor n/2 \rfloor$. Daniel Lemire asked whether such a polynomial exists in every degree. We translate the condition into a question about one ray class and then count that class with the polynomial von Mangoldt function. Most of the resulting character sum can now be bounded. The part we cannot bound is a signed sum over characters of high 2-power order. We state the needed bound and prove that it would finish the argument. Separate certified tests cover every degree through 3000. A companion document records what has been tried, what is provably blocked, and what might still work.

1 The argument

Put $\ell = \lceil n/2 \rceil - 1$. Figure 1 shows what is proved and what is not.

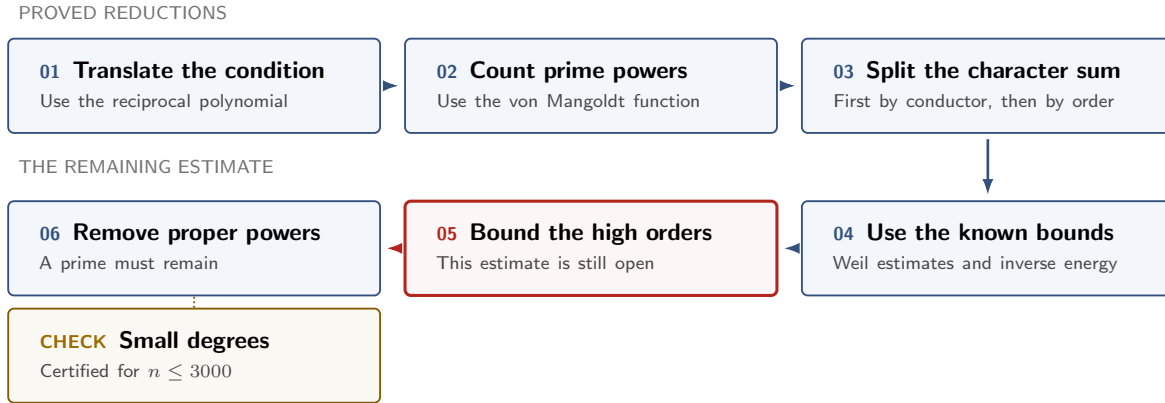


Figure 1: Steps 1–4 and 6 are proved. Step 5 is open. The gold box records the separate computation for $n \leq 3000$.

2 The reduction (**PROVED**)

1. Translate the condition. For $j \geq 0$ set $\mathcal{E}_j = (1 + x\mathbb{F}_2[x])/(x^{j+1})$, $|\mathcal{E}_j| = 2^j$, and for monic F let $\langle F \rangle_j = x^{\deg F} F(x^{-1}) \bmod x^{j+1}$. Then $\deg(f - x^n) \leq \lfloor n/2 \rfloor$ if and only if $\langle f \rangle_\ell = 1$: the conjecture asks whether one ray class contains a prime of degree n . Reversing f gives the arithmetic-progression form — a prime P of degree n with $P \equiv 1 \pmod{x^{\lceil n/2 \rceil}}$ — so this is Linnik's problem with exponent exactly 2 at a prime-power modulus, equivalently Legendre's conjecture for $\mathbb{F}_2[t]$. In Witt-vector terms (Katz's splitting $\mathcal{E}_j \cong \prod_{k \text{ odd} \leq j} \mathbb{Z}/2^{e_k}$, $e_k = \lfloor \log_2(j/k) \rfloor + 1$) the identity class is the locus where all odd-power Galois-ring traces vanish to these dyadic precisions.

2. Count prime powers. With $N_j(b) = \sum_{\deg F=n, \langle F \rangle_j=b} \Lambda(F)$, logarithmic differentiation gives this weighted count exactly. At the odd endpoint $n = 2\ell + 1$, $N_\ell(1) = 1 + nI_n(1)$; at $n = 2\ell + 2$ odd proper powers do not meet the identity class and a direct count bounds the rest. Either way it suffices to prove

$$N_\ell(1) > 2^{n-\ell} - 2^\ell. \quad (1)$$

3. Split by conductor. Each $\mathcal{E}_j \rightarrow \mathcal{E}_{j-1}$ has two-element fibres; writing $H_j(b)$ for the difference of the two child populations above b , repeated binary splitting telescopes to

$$2^\ell N_\ell(e) = 2^n + \sum_{j=1}^{\ell} \varepsilon_j(e) 2^{j-1} H_j(b_j(e)), \quad \varepsilon_j(e) \in \{\pm 1\}, \quad (2)$$

and for the primitive characters X_j of conductor j , $H_j(b) = 2^{1-j} \sum_{\chi \in X_j} \overline{\chi(b)} S_n(\chi)$ with $|H_j(b)| \leq (j-1)2^{\lceil n/2 \rceil}$ by the function-field Riemann hypothesis. That is strong enough at small conductors; applied separately at every conductor it overshoots by a factor of order ℓ . *The whole problem is that factor.*

4. A characteristic-two bilinear bound. The inverse-energy estimate that Type-I/II arguments need does hold in characteristic two, including products wrapping modulo x^r : group collisions by x -adic valuation, turn each group into a bounded-degree factorisation problem by a Padé approximation, then use lift counts and divisor bounds. It does not finish the problem, because applying it one term at a time destroys the signs in (2). (The published Type-I/II theorems of this shape assume odd characteristic; see §5.)

5. Add the large conductors before taking absolute values. Let $c = \lceil \log_2 \ell \rceil$, $a = \ell - c - 1$, $W_{\ell,n} = 2^{\lceil n/2 \rceil} \sum_{1 \leq j < a} (j-1)2^{j-1}$ and $B_{\ell,n} = 2^{2\ell} - W_{\ell,n} > 0$. For the identity class the terms telescope:

$$C_{\ell,n} := \sum_{j=a}^{\ell} 2^{j-1} H_j(1) = 2^{\ell} N_{\ell}(1) - 2^{a-1} N_{a-1}(1), \quad (3)$$

and it is enough to prove $C_{\ell,n} > -B_{\ell,n}$, which we call (REL): the small conductors contribute at least $-W$, so (REL) gives $2^{\ell} N_{\ell}(1) - 2^n > -W - B = -2^{2\ell}$, which is (1).

6. Split the remaining characters by order. For $s \geq 0$ put $h_{j,s} = 2^{j-\lfloor j/2^s \rfloor}$ and $P_{j,s} = \sum_{g \in 2^s \mathcal{E}_j} N_j(g)$; orthogonality gives the sum over characters of conductor j and exact order 2^s as $T_{j,s} = h_{j,s} P_{j,s} - h_{j,s-1} P_{j,s-1} - h_{j-1,s} P_{j-1,s} + h_{j-1,s-1} P_{j-1,s-1}$. With Q the largest power of two such that $3Q \lceil \log_2 \ell \rceil \leq \ell$, the Weil bound handles every order at most Q , leaving $O(\log \log \ell)$ high orders per conductor — at $\ell = 200$, 47 of the 67 conductor–order pairs.

3 What remains open

7. Bound the high orders (OPEN). For $a \leq j \leq \ell$ and $2^s > Q$ with the corresponding character set nonempty, the following would be enough:

$$4\ell |T_{j,s}(n)| \leq \#X_{j,s}(j-1)2^{\lceil n/2 \rceil}. \quad (\text{HWO})$$

Equivalently, without characters: with $\Delta_{j,s} = 2P_{j,s} - P_{j-1,s}$, $d_s = \lfloor (j-1)/2^s \rfloor$ and $R_{j,s} = 2^{d_s-1-d_s}$, (HWO) is $4\ell |R_{j,s} \Delta_{j,s} - \Delta_{j,s-1}| \leq (R_{j,s} - 1)(j-1)2^{\lceil n/2 \rceil}$ when $2^{s-1} \nmid j$, and the direct sparse-discrepancy bound $4\ell |\Delta_{j,s}| \leq (j-1)2^{\lceil n/2 \rceil}$ when $2^{s-1} \mid j$ but $2^s \nmid j$. So a Witt-precision comparison addresses only the nonresonant layers, and resonant ones occur in the target range ($j = 200$, $2^s = 16$). Then

$$\boxed{(\text{HWO}), \text{ or another proof of (REL)}} \implies (\text{REL}) \implies (1) \implies \text{a prime in the identity class.}$$

A proof of (HWO) for $\ell \geq 200$ and $n \in \{2\ell+1, 2\ell+2\}$ proves the conjecture for all n ; every degree $n \leq 3000$ carries an exhibited, doubly certified half-degree irreducible (**FINITE**).

The requirement is a dial, and it is smaller than 4ℓ . By orthogonality $N_{\ell}(1) > 0$ exactly when $|\sum_{\chi \neq 1} S_n(\chi)| < 2^n - 2^{\ell} \Theta$, Θ the proper-power mass, while Weil bounds the left side by $2^{n/2} D(\ell)$ with $D(\ell) = \sum_{j \leq \ell} (j-1)2^{j-1} = (\ell-2)2^{\ell} + 2$. So the whole problem is one aggregate saving: if that sum is at most $2^{n/2} D(\ell)/F$, one gets an irreducible with $\deg(f-x^n) \leq \lfloor n/2 \rfloor + k$ for $k = \log_2((\ell-2)/F) + O(1)$. Here $F = 1$ is the known unconditional bound (Hayes; sharp form Hsu, Cohen, Gao), giving $k = \log_2 n - 1 + O(1)$, and $k = 0$ is the conjecture — so *doubling F buys exactly one coefficient*, and any unconditional $F > 1$ at $q = 2$ would already be new. In this aggregate form the requirement at $\ell = 200$ is $F > 140.007$ (n odd) and $F > 99.000$ (n even), that is $F > (\ell-2)/\kappa$ with $\kappa = 2^{n/2-\ell}$, rather than the $4\ell = 800$ of (HWO): the chain's extra factor is mostly its use of the weaker $B = 2^{2\ell} - W$ in place of $2^n - 2^{\ell} \Theta - W$. The aggregate statement also dispenses with a, c, Q, W and B altogether and lets one layer subsidise another, which the exact data show happening by two orders of magnitude — but it does not change the shape of what must be proved.

4 Two sufficient statements, and what is measured

(CYL), the smallest known. Let $K = \ker(\mathcal{E}_{\ell} \rightarrow \mathcal{E}_{a-1})$, elementary abelian of order $2^{\ell-a+1}$, and for a character ψ of K let $A_{\psi} = \sum_{\deg F=n, \langle F \rangle_{a-1}=1} \Lambda(F) \psi(\langle F \rangle_{\ell})$ be the prime mass of a cylinder twisted by a sign pattern of the coefficients of $x^{n-a}, \dots, x^{n-\ell}$. Fourier inversion on K and $B_{\ell,n} \geq 2^{2\ell-1}$ give: $|A_{\psi}| < 2^{\ell-1}$ for every $\psi \neq 1$ implies (REL). Weil gives only $|A_{\psi}| \leq (\ell-1)2^{\lceil n/2 \rceil}$, a factor $4(\ell-1)$ short; the random model predicts $|A_{\psi}| \approx 2^{(n-a+1)/2} \sqrt{\ell}$. Exact computation for $12 \leq \ell \leq 24$ at both endpoints matches that prediction within 8% and finds (CYL) true from $\ell = 16$ (n odd) and $\ell = 18$ (n even) on, $\max_{\psi} |A_{\psi}|/2^{\ell-1}$ falling to 0.09 and 0.15 at $\ell = 24$, and likewise for all 2^{a-1} translated cylinders from $\ell = 20$. This is finite evidence, not a proof. It does resist the one disproof route available at fixed q : since $|K| < 8\ell$, a Plancherel

forcing argument gains at most $\sqrt{8\ell}$ against a root-mean-square that is $\Theta(\ell 2^{-\ell/2})$ of the threshold, and no A_ψ vanishes at any computed endpoint (for $\ell = 23$, $n = 47$ the total mass is odd, which forces it).

The geometric face, re-posed. By Katz the characters of exact conductor j are the $\mathbb{F}_2$ -points of $\text{Prim}_j \cong \mathbb{G}_m \times \mathbb{A}^{j-1}$, carrying a lisse L_{univ} of rank $j - 1$ whose Frobenius traces are the $S_n(\chi)$; the conductor- j sum is a Lefschetz trace of the Adams twist $\Xi_n(L_{\text{univ}})$. Deligne’s weight bounds show that a bound on the sum of Betti numbers *alone* can never give (HWO): with the top cohomology present one would need a Betti sum below $1/4$. What is needed is a *degree* statement — the top $\approx 6 + 2 \log_2 C$ cohomology groups must vanish — together with a Betti bound $\lesssim 2^{j/2}$. Two facts narrow this. Geometric monodromy contains $SL(j - 1)$ for every $j \geq 4$ in characteristic two (Katz, IMRN 2013, Thm. 5.1; $j = 3$ is finite, Gorodetsky 2019), which forces $H_c^{2j} = 0$ for $j \geq 4$; and an exact L -function computation over $\mathbb{F}_{2^r}$ ($j \leq 7$) puts the top weight at $n + j + O(1)$, never $n + 2j - O(1)$, matching the $\mathbb{F}_2$ layer sums at $\ell = 24$. The residual gap is the characteristic-two case of Sawin’s Hypothesis H (arXiv:1810.01303), whose only unconditional input is vacuous at $p = 2$, plus a uniform Betti bound ($2^{O(j \log j)}$ is the best known).

5 What is provably blocked

Four barriers are proved, and they explain why the factor ℓ survives every elementary attack. *Moduli-only*: an explicit nonnegative fake population has the true low-conductor Fourier data, all high moduli inside Weil, second moment below the truth, and an empty identity class. *Symmetry*: every degree-preserving symmetry lies in $PGL_2(\mathbb{F}_2)$, only the Borel descends to $\mathcal{E}_\ell$, and the orbit of the identity class under all structural symmetries has size at most 2. *Construction*: provably irreducibility-preserving constructions multiply the degree, so window families are lacunary and none ever reaches a prime n — though monomial substitution $f \mapsto f(x^t)$ never leaves the window ($f(x^t) - x^{mt} = (f - x^m)(x^t)$), so by Lidl–Niederreiter Thm. 3.35 every in-window seed of degree m and order e gives in-window irreducibles of degree mt whenever every prime of t divides e but not $(2^m - 1)/e$; from the certified seeds this reaches 9.3% of composite $n \leq 10^5$, and the case $m = 2$ (Jeřábek, 2011) is $n = 2 \cdot 3^k$. *Parity*: the window has exact level of distribution $|W_n|$ with zero remainder, which yields an element with all irreducible factors of degree $> (\frac{1}{4} - \varepsilon)n$ and, with Kuhn weights, one with at most three such factors — but for $10 \leq n \leq 15$ there is an explicit nonnegative rational population with the window’s exact divisor data and no irreducible in its support, so no lower-bound sieve at that level can produce a prime; and any such certificate applies to every window, hence would prove Legendre for $\mathbb{F}_2[t]$.

Over $\mathbb{F}_q$ the conjecture is a theorem once q is large: $q > (n + 1)^2/4$ suffices (Hsu; Cohen), and $q > 7101p^2$ suffices for all large n when p is odd (Bagshaw, arXiv:2401.10399, Cor. 2.5, via the reversal duality) — a threshold independent of n , but one that excludes characteristic two, every prime field, and every $q = p^2$. The conjecture is open precisely because $q = 2$ is fixed.

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